

Input artefacts · Stages 1–2b

Per-function feature blocks

Assembly and output

Dalvik bytecode of v

invoke-virtual {v0}, L...
const-string v1, "..."
move-result-object v2
from classes*.dex · Stage 1

Function-call graph $G = (V, E)$

 v

Suspicious-API tags and backward slices

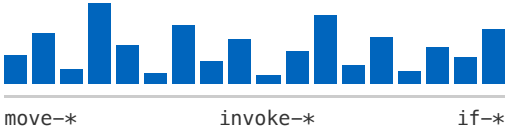
MobSF tagging · dangerous permissions — Stage 2a
Sliced CFGs at each call site — Stage 2b

Method body of v

Syntactic paths through the decompiled method, encoded by a pre-trained code2vec model

3.1 Opcode histogram hop

Frequency distribution of the Dalvik instructions in the body of v — raw one-hot counts, or TF-IDF weighted against the whole corpus to surface discriminative opcodes.



|0|

3.2 Structural metrics $sstruct$

Position and influence of v inside the FCG. Standardised so that no metric dominates the learning process through its numerical range.

In-degree

Out-degree

PageRank

Betweenness

4

3.3 API flags and slice counts $aapi$

Direct, method-level indicators of the access-to-transmit exfiltration pattern: two binary flags plus the number of sensitive calls appearing in the slices associated with v .

has_access1

has_transmit1

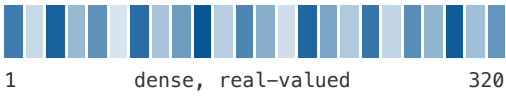
#access3

#transmit2

4

3.4 Semantic embedding $c2v$

A dense 320-dimensional code2vec vector carrying the contextual meaning of the method — costlier than handcrafted features, but it recovers similarity across renamed, reused and repackaged code.



320

Concatenate

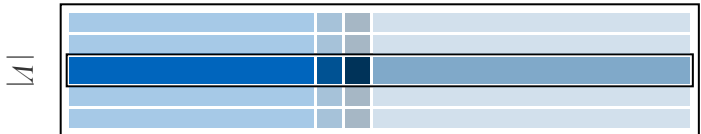


$x_v = [hop \parallel sstruct \parallel aapi \parallel c2v] \in \mathbb{R}^F$

▼ z-score normalisation, per feature column

$\hat{x} = (x - \mu) / \sigma$ comparable scales across heterogeneous blocks

Node-feature matrix



$X \in \mathbb{R}^{|V| \times F}$

$F = |0| + 4 + 4 + 320$

Output

Serialised with the unified hyperedge set by the Hypergraph Constructor, Stage 6 — $\{H, W, X\}$.

The code2vec block is reused in **Stage 4B** for the semantic k -nearest-neighbour hyperedges ($\tau \approx 0.80$, $k = 5$).

Access and transmit evidence also weights the slice-aware hyperedges of **Stage 4C**.